INCH-POUND
MIL-PRF-3098/71E
21 April 2009
SUPERSEDING
MIL-PRF-3098/71D
17 May 2004

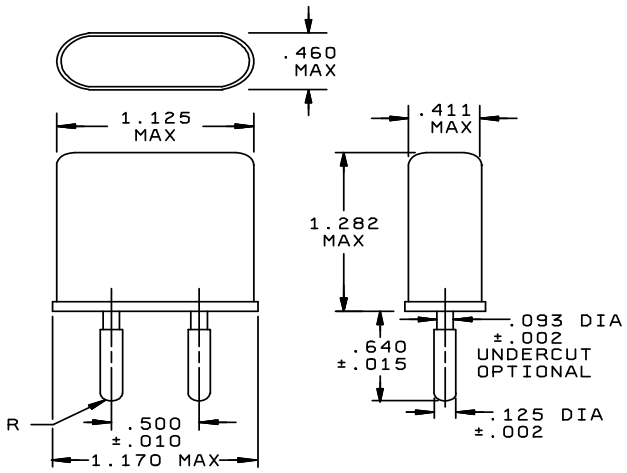
PERFORMANCE SPECIFICATION SHEET

CRYSTAL UNIT, QUARTZ, CR96/U

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-3098.

Pertinent characteristics: 2.0 MHz to 9.25 MHz; fundamental; noncontrolled; antiresonance.



Inches	mm	Inches	mm	Inches	mm
.002	.05	.125	3.18	.640	16.26
.010	.25	.411	10.44	1.125	28.58
.015	.38	.460	11.68	1.170	29.72
.093	2.36	.500	12.70	1.282	32.56

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Marking to be in accordance with MIL-PRF-3098.

FIGURE 1. Crystal unit - CR96/U.

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REQUIREMENTS:

Dimensions, marking, and configuration: See figure 1.

Frequency range: 2.0 MHz to 9.25 MHz, inclusive.

Mode of oscillation: Fundamental.

Rated drive level: 1.0 mW, maximum.

Antiresonance, load capacitance: 35.0 pF ±0.5 pF.

Operating temperature range (noncontrolled): -35°C to +75°C, inclusive.

Frequency tolerance: ±50 parts per million (ppm).

Equivalent resistance: See table I.

Shock:

Frequency change permitted: ± 5 ppm.

Equivalent resistance change permitted: ±15 percent, or 2 ohms, whichever is greater.

Vibration: Method 201 of MIL-STD-202.

Frequency change permitted: ± 5 ppm.

Equivalent resistance change permitted: ±15 percent, or 2 ohms, whichever is greater.

Temperature cycling:

Frequency change permitted: ±5 ppm.

Equivalent resistance change permitted: ±15 percent, or 2 ohms, whichever is greater.

Aging:

Frequency change permitted: ±5 ppm.

TABLE I. Equivalent resistance.

Frequency range, inclusive	Maximum resistance
MHz	Ohms
2.00 to 4.75	120
4.75+ to 6.00	75
6.00+ to 7.50	50
7.50+ to 10.00	35

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Referenced documents. In addition to MIL-PRF-3098, this document references the following:

MIL-STD-202

The margins of this specification sheet are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - CR
Navy - EC
Air Force - 99
DLA - CC

Review activities:
Air Force - 19, 84

Preparing activity:
DLA - CC

(Project 5955-2009-005)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.

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